

OpenStax Calculus Volume 3

Section 7.4 Exercises: Series Solutions of Differential Equations

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Find a power series solution for the following differential equations.

1. $y'' + 6y' = 0$
2. $5y'' + y' = 0$
3. $y'' + 25y = 0$
4. $y'' - y = 0$
5. $2y' + y = 0$
6. $y' - 2xy = 0$
7. $(x - 7)y' + 2y = 0$
8. $y'' - xy' - y = 0$
9. $(1 + x^2)y'' - 4xy' + 6y = 0$
10. $x^2y'' - xy' - 3y = 0$
11. $y'' - 8y' = 0$, $y(0) = -2$, and $y'(0) = 10$
12. $y'' - 2xy = 0$, $y(0) = 1$, and $y'(0) = -3$
13. The differential equation $x^2y'' + xy' + (x^2 - 1)y = 0$ is a Bessel equation of order 1. Use a power series of the form $y = \sum_{n=0}^{\infty} a_n x^n$ to find the solution.